

INTEL PHYSICAL AI ONLINE CHALLENGE · AI INFRA SUMMIT 2026

TwinGuard

Closed-Loop Bimanual Physical AI with Coordinated Handover, Intel Anomalib Visual Guard & OpenVINO INT8 Acceleration

Intel Physical AI Studio

OpenVINO 2025

NNCF INT8 Quantized

Anomalib v2.6 Visual Guard

MuJoCo 3.1+ RK4

Dual SO-101 (12-DOF)

112 / 112 Tests Passed

Team TwinGuard · Platform: lablab.ai

Why Open-Loop Physical AI Fails

✗ The Problem: Open-Loop Blindness

Current Vision-Language-Action (VLA) foundation models generate motor trajectories without continuous verification:

- ⚠️ **Unmodeled Physics:** Grasp slippage, friction variations, or stuck drawers cause immediate failure.
- ⚠️ **Zero Error Recovery:** Open-loop controllers continue moving through empty air, dropping dishes and causing catastrophic collisions.
- ⚠️ **Rigid Arm Isolation:** Dual-arm setups usually operate as two single arms rather than coordinated bimanual partners.

✓ The TwinGuard Solution: Closed-Loop AI

A hierarchical Physical AI architecture combining multimodal planning with self-healing:

- 🛡️ **Dual-Layer Verifier:** SafetyVerifier checks contact normal forces and drop limits; VisualGuard checks visual anomalies.
- 🔄 **Dynamic Self-Healing:** Automated re-observation and replanning that turns a failed grasp into a successful recovery.
- 🤝 **True Bimanual Handover:** Mid-air object transfer between arms with synchronized clearance.

End-to-End System Architecture



Intel Core Ultra Workload Placement

COMPUTE ELEMENT	PHYSICAL AI ROLE	WORKLOAD & OPTIMIZATION	MEASURED PERFORMANCE
Intel NPU AI Boost Neural Engine	Low-power continuous vision perception & anomaly embedding	OpenVINO INT8 (NNCF Quantized) : Object detector and Anomalib visual guard models executing in ultra-low power budget.	0.58 ms latency ~1,700 FPS throughput
Intel Arc iGPU Integrated Graphics	High-bandwidth camera rendering & frame pipeline	OpenVINO Async Mode (start_async()) : Parallel tensor transformations, multi-view camera ingestion (overview + wrist).	1,439 FPS Async pipeline throughput
Intel CPU Performance & Efficient Cores	Physics dynamics, kinematics & safety verifier	MuJoCo 500 Hz RK4 Engine : Damped Least Squares IK, contact normal force accumulation, collision checking, and LLM reasoning.	5.3x Real-Time Speedup over wall-clock
Dynamic Hetero Auto Dispatch	Automated multi-device load balancing	OpenVINO --device AUTO / HETERO:NPU,CPU : Seamless runtime failover and hardware concurrency.	Zero Dropped Frames Deterministic execution

Autonomous Failure Diagnosis & Recovery

Controlled Failure Injection: Seed 2

To empirically validate closed-loop self-healing under the challenge rubric, physical grasp slips were deliberately injected into **Seed 2** and **Seed 7**:

[VERIFIER WARNING] Step 7 [grasp] failed verification:
Contact normal force lost & object freejoint escaped grasp zone (dist 0.18m > tol 0.15m).

[AUTONOMOUS RECOVERY] Re-observing scene state...
Dynamic replanner synthesized recovery approach. Second grasp succeeded. Bimanual pour completed.

100%

POST-RECOVERY TASK COMPLETION RATE (10/10 SEEDS)

0

ARM-TO-ARM OR TABLE COLLISIONS ACROSS 100+ STEPS

< 1.0s

AVERAGE WALL-CLOCK EXECUTION TIME PER FULL TASK

OpenVINO Benchmarks & Test Suite

OpenVINO INT8 vs FP32 Comparison

METRIC	FP32 BASELINE	NNCF INT8 QUANTIZED
Weights Size	723 KB	363 KB (1.99x smaller)
Mean Latency	0.834 ms	0.587 ms (1.42x faster)
Sync Throughput	1,198 FPS	1,703.8 FPS
Async Throughput	1,266 FPS	1,439.2 FPS
Visual Guard Latency	0.836 ms	1,196.3 FPS (Anomalib)

Production Engineering Hygiene

- ✓ **112 / 112 Pytest Tests Passed:** 100% test pass covering kinematics, verifiers, planners, and VLA schemas.
- ✓ **Zero Hardcoded Magic Numbers:** Centralized configuration in `configs/sim_config.yaml`.
- ✓ **Single-Command Docker:** Fully reproducible Linux container build (`docker build -t twinguard`).
- ✓ **Apache-2.0 Licensed:** Complete attribution for SO-ARM100 and LeRobot upstream projects.

Ready for the Real World

TwinGuard brings closed-loop certainty to bimanual Physical AI.

By coupling multimodal reasoning with continuous physical verification and Intel OpenVINO edge acceleration, TwinGuard transforms robots from fragile open-loop actuators into autonomous, self-healing manipulators.

100%

Task Success Across 10 Seeds

< 1 ms

OpenVINO INT8 Latency

112

Verified Unit Tests